Fall 2025, Carnegie Mellon University

# Cognition in the Age of AI (85-414/85-814)

**Instructor:**
Dr. Maggie Henderson (she/her)
Assistant Professor in Psychology & Neuroscience Institute
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**Course meeting times:** MW 11am – 12:20pm, BH 340A (Baker Hall)
**Office Hours:** Mondays, 3-5 pm (Baker Hall 335G)
**Course website:** https://canvas.cmu.edu/courses/48790
**Course GitHub page:** https://github.com/hendersonneurolab/CogAI_Fall2025

**Course description:**
Advances in artificial intelligence and internet-scale datasets have transformed our everyday lives. In this course, we will learn about ways that AI and large datasets are also transforming what we know about the human mind and brain. We will discuss recent research that uses tools like deep convolutional neural networks and large language models as "model systems" for investigating human cognition, and what we can learn by comparing model outputs against real human data. We will also examine the rise of big datasets in neuroscience and psychology research, and how these datasets are enabling powerful applications of machine learning to the study of human cognition. Compared to traditional approaches, these advances are making it possible to conduct more "naturalistic" psychology experiments, where tasks and stimuli more closely match the complexity of our real-world experiences. Throughout the course, we will consider how these tools are opening up progress on questions that have historically been challenging to study, such as how perception unfolds within complex natural environments, how we comprehend language, and how we learn about the world through experience. The course format will involve weekly readings and discussion.

**Course learning goals:**

After finishing the course, students should be able to:

- Articulate the key recent advances in cognitive neuroscience research that have come from AI and the use of large-scale datasets.
- Critically examine claims of similarities and/or differences between AI systems and humans, as they relate to perception and cognition.
- Understand the basics of modern deep neural networks, in terms of their architectures, datasets, training procedures, and evaluation, and how these elements are implemented in Python code.
- Understand methods for human brain recording, and compare the different approaches for computational modeling of human brain data.
- Read and explain journal articles that use AI and ML as tools for neuroscience.

**Course schedule:**

<table>
<tr><th>Week 1</th><th colspan="4">Introduction to the course</th></tr>
<tr><td>Monday:</td><td>8/25</td><td>Wednesday:</td><td>8/27</td><td rowspan="8">Readings:<br>No required readings this week - see "Further readings" on Canvas for optional resources.</td></tr>
<tr><td colspan="4">Due dates:</td></tr>
<tr><td colspan="3">Reading reflection due:</td><td></td></tr>
<tr><td colspan="3">Lab exercise (from last week) due:</td><td></td></tr>
<tr><td colspan="3">Self evaluations due:</td><td></td></tr>
<tr><td colspan="3"></td><td></td></tr>
<tr><td colspan="4">Other notes:</td></tr>
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<tr><th>Week 2</th><th colspan="4">Convolutional neural networks (CNNs) and human visual processing</th></tr>
<tr><td>~~Monday:~~</td><td>~~9/1~~</td><td>Wednesday:</td><td>9/3</td><td rowspan="8">Readings:<br>1. Lindsay (2021). Convolutional Neural Networks as a Model of the Visual System: Past, Present, and Future. https://doi.org/10.1162/jocn_a_01544</td></tr>
<tr><td colspan="4">Due dates:</td></tr>
<tr><td colspan="3">Reading reflection due:</td><td>9/2</td></tr>
<tr><td colspan="3">Lab exercise (from last week) due:</td><td></td></tr>
<tr><td colspan="3">Self evaluations due:</td><td>9/3</td></tr>
<tr><td colspan="3"></td><td></td></tr>
<tr><td colspan="4">Other notes:<br>Monday is Labor Day, no class</td></tr>
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<table>
<tr><th>Week 3</th><th colspan="4">How similar are CNNs and humans? Texture and shape bias</th></tr>
<tr><td>Monday:</td><td>9/8</td><td>Wednesday:</td><td>9/10</td><td rowspan="8">Readings:<br>1. Geirhos et al. (2019). ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness. https://arxiv.org/abs/1811.12231<br>2. Firestone (2020). Performance vs. competence in human-machine comparisons. https://doi.org/10.1073/pnas.1905334117<br>3. (optional) Tartaglini et al. (2022). A Developmentally-Inspired Examination of Shape versus Texture Bias in Machines. https://arxiv.org/abs/2202.08340</td></tr>
<tr><td colspan="4">Due dates:</td></tr>
<tr><td colspan="3">Reading reflection due:</td><td>9/7</td></tr>
<tr><td colspan="3">Lab exercise (from last week) due:</td><td>9/10</td></tr>
<tr><td colspan="3">Self evaluations due:</td><td>9/10</td></tr>
<tr><td colspan="3"></td><td></td></tr>
<tr><td colspan="4">Other notes:</td></tr>
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| Week 4 | Other modalities: Language and multi-modal models | | | |
|---|---|---|---|---|
| Monday: | 9/15 | Wednesday: | 9/17 | **Readings:**<br>1. Schulze Buschoff et al. (2025). Visual cognition in multimodal large language models. https://doi.org/10.1038/s42256-024-00963-y<br>2. Anthropic (2024). Mapping the Mind of a Large Language Model. https://www.anthropic.com/research/mapping-mind-language-model |
| **Due dates:** | | | | |
| | | Reading reflection due: | **9/14** | |
| | | Lab exercise (from last week) due: | **9/17** | |
| | | Self evaluations due: | **9/17** | |
| Other notes: | | | | |

| Week 5 | Adversarial examples and peering into the "black box" | | | |
|---|---|---|---|---|
| Monday: | 9/22 | Wednesday: | 9/24 | **Readings:**<br>1. Veerabadran et al. (2023). Subtle adversarial image manipulations influence both human and machine perception. https://doi.org/10.1038/s41467-023-40499-0<br>2. Olah et al. (2017). Feature Visualization. https://distill.pub/2017/feature-visualization/ |
| **Due dates:** | | | | |
| | | Reading reflection due: | **9/21** | |
| | | Lab exercise (from last week) due: | **9/24** | |
| | | Self evaluations due: | **9/24** | |
| Other notes: | | | | |

| Week 6 | Large-scale recording from the brain: fMRI and other methods | | | |
|---|---|---|---|---|
| Monday: | 9/29 | Wednesday: | 10/1 | **Readings:**<br>1. Allen et al. (2022). A massive 7T fMRI dataset to bridge cognitive neuroscience and artificial intelligence. **Only highlighted regions are required.** https://doi.org/10.1038/s41593-021-00962-x<br>2. Lindquist (2020). Pipeline choices alter neuroimaging findings. https://www.nature.com/articles/d41586-020-01282-z |
| **Due dates:** | | | | |
| | | Reading reflection due: | **9/28** | |
| | | Lab exercise (from last week) due: | **10/1** | |
| | | Self evaluations due: | **10/1** | |
| Other notes: | | | | |

| **Week 7** | **How do we compare neural networks and the brain?** |
|---|---|
| Monday: **10/6** Wednesday: **10/8** | **Readings:**<br>1. Yamins et al. (2014). Performance-optimized hierarchical models predict neural responses in higher visual cortex. https://doi.org/10.1073/pnas.1403112111<br>2. Naselaris et al. (2012). Encoding and decoding in fMRI. https://doi.org/10.1016/j.neuroimage.2010.07.073. **Only highlighted regions are required.** |
| **Due dates:** | |
| Reading reflection due: **10/5** | |
| Lab exercise (from last week) due: **10/8** | |
| Self evaluations due: **10/8** | |
| Other notes:<br>**Fall Break next week!** | |

| **Week 8** | **Which neural networks are best at predicting the visual system?** |
|---|---|
| Monday: **10/20** Wednesday: **10/22** | **Readings:**<br>1. Güçlü & van Gerven (2015). Deep Neural Networks Reveal a Gradient in the Complexity of Neural Representations across the Ventral Stream. https://doi.org/10.1523/JNEUROSCI.5023-14.2015<br>2. Conwell et al. (2024). A large-scale examination of inductive biases shaping high-level visual representation in brains and machines. https://doi.org/10.1038/s41467-024-53147-y |
| **Due dates:** | |
| Reading reflection due: **10/19** | |
| Lab exercise (from last week) due: **10/22** | |
| Self evaluations due: **10/22** | |
| **Final Project Idea Survey:** **10/22** | |
| Other notes: | |

| **Week 9** | **Neural decoding: how close are we to "mind-reading"?** |
|---|---|
| Monday: **10/27** Wednesday: **10/29** | **Readings:**<br>1. Kay et al. (2008). Identifying natural images from human brain activity. https://doi.org/10.1038/nature06713<br>2. Tang et al. (2023). Semantic reconstruction of continuous language from non-invasive brain recordings. https://doi.org/10.1038/s41593-023-01304-9 |
| **Due dates:** | |
| Reading reflection due: **10/26** | |
| Lab exercise (from last week) due: **10/29** | |
| Self evaluations due: **10/29** | |
| Other notes: | |

| Week 10 | Can neural networks be used to model human development? |
|---|---|
| Monday: **11/3** Wednesday: **11/5** | **Readings:**<br>1. Jinsi*, Henderson*, et al. (2023). Early experience with low-pass filtered images facilitates visual category learning in a neural network model. https://doi.org/10.1371/journal.pone.0280145<br>2. Orhan & Lake (2024). Learning high-level visual representations from a child's perspective without strong inductive biases. https://doi.org/10.1038/s42256-024-00802-0 |
| **Due dates:** | |
| Reading reflection due: **11/2** | |
| Lab exercise (from last week) due: **11/5** | |
| Self evaluations due: **11/5** | |
| **Final Project Proposal: 11/5** | |
| Other notes: | |

| Week 11 | Using generative image and video models for neuroscience |
|---|---|
| Monday: **11/10** Wednesday: **11/12** | **Readings:**<br>1. Luo et al. (2023). Brain Diffusion for Visual Exploration: Cortical Discovery using Large Scale Generative Models. https://www.cs.cmu.edu/~afluo/BrainDiVE/<br>2.Yeung et al. (2025). Reanimating images using neural representations of dynamic stimuli. https://brain-nrds.github.io/ |
| **Due dates:** | |
| Reading reflection due: **11/9** | |
| Lab exercise (from last week) due: **11/12** | |
| Self evaluations due: **11/12** | |
| Other notes: | |

| Week 12 | Neuro-to-AI Transfer: can brain data improve neural network performance? |
|---|---|
| Monday: **11/17** Wednesday: **11/19** | **Readings:**<br>1. Aquino et al. (2025). Brain2Model Transfer: Training sensory and decision models with human neural activity as a teacher. https://arxiv.org/pdf/2506.20834 |
| **Due dates:** | |
| Reading reflection due: **11/16** | |
| Lab exercise (from last week) due: | |
| Self evaluations due: **11/19** | |
| Other notes: | |

<table>
<tr><td>Week 13</td><td colspan="4">Final presentations</td></tr>
<tr><td>Monday:</td><td>11/24</td><td>~~Wednesday:~~</td><td>~~11/26~~</td><td rowspan="7">Readings:<br>None</td></tr>
<tr><td colspan="4">Due dates:</td></tr>
<tr><td colspan="3">Reading reflection due:</td><td></td></tr>
<tr><td colspan="3">Lab exercise (from last week) due:</td><td></td></tr>
<tr><td colspan="3">Self evaluations due:</td><td></td></tr>
<tr><td colspan="3"></td><td></td></tr>
<tr><td colspan="4">Other notes:<br>Thanksgiving on Wednesday, no class!<br>See Canvas for presentation schedule.</td></tr>
<tr><td>Week 14</td><td colspan="4">Final presentations</td></tr>
<tr><td>Monday:</td><td>12/1</td><td>Wednesday:</td><td>12/3</td><td rowspan="7">Readings:<br>None</td></tr>
<tr><td colspan="4">Due dates:</td></tr>
<tr><td colspan="3">Reading reflection due:</td><td></td></tr>
<tr><td colspan="3">Lab exercise (from last week) due:</td><td></td></tr>
<tr><td colspan="3">Self evaluations due:</td><td></td></tr>
<tr><td colspan="3">Final paper:</td><td>12/10</td></tr>
<tr><td colspan="4">Other notes:<br>Final paper is due December 10th!</td></tr>
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## Evaluation & Grading:

| | |
|---|---|
| 30% | Weekly reading reflections |
| 25% | Weekly lab assignments |
| 10% | Attendance & participation |
| 5% | Final project proposal |
| 15% | Final project write-up |
| 15% | Final project presentation |

Reading reflections (30% of grade):

- Each week, you'll have 1-2 assigned readings. You will be responsible for submitting a weekly reading reflection on Canvas, including a list of discussion questions and any points of confusion.
- Estimated length for the reflections is around half a page of text.
- See the course Canvas site for more detailed instructions.
- Reflections will be due each week by the **Sunday evening before class (11:59 pm).**
- During the semester, you may miss one reading reflection without any penalty.
  - Equivalently, this means the lowest score is dropped.

Lab assignments (25% of grade):

- Each week we will have a hands-on coding assignment to accompany the readings.
- Lab assignments will use Python, and will be structured as tutorial notebooks in Google Colab. You'll submit your finished assignments on Canvas.
- Lab assignments can all be found on the class GitHub page, at:
  - https://github.com/hendersonneurolab/CogAI_Fall2025
- We will devote some in-class time to starting work on the lab assignments, and you'll complete the rest of each assignment on your own.
- Lab assignments for each week will be due the following **Wednesday evening (11:59 pm).**
  - Not the same day you receive it, but the following week.
- During the semester, you may miss one lab assignment without any penalty.
  - Equivalently, this means the lowest score is dropped.

Attendance & participation (10% of grade):

- This course involves class discussion, so your presence and participation is critical for all of our success.
- Participation will be graded using a self-evaluation form on Canvas.
  - *As long as you are present in class and fill out the self-evaluation form completely, you will receive full points for attendance / participation.*
- Self-evaluation forms for each week (for both Monday and Wednesday sessions) will be due each **Wednesday evening (11:59 pm).**
  - *If you're not in class on a particular day, don't fill out this form.*
- During the semester, you may miss up to 2 class sessions (or self-evaluation forms) without penalty. If you have extenuating circumstances and need to miss more than 2 sessions, please email me and we can work on a solution.

Final project (35% of grade):

- Your final project for the course will be to analyze a recent paper from the research literature that uses AI, ML, or data science tools to address a topic in psychology or neuroscience research.
- The grade for this project is split across several components:
  - 5%: Provide an initial "proposal" for your project idea.

        - Due **November 5th (11:59 pm)**
    - 15%: Give a 15 minute presentation in class discussing the paper.
        - During class sessions between **November 24th – December 3rd.**
    - 15%: Write a 3-4 page report discussing your selected paper.
        - Due **December 10th (11:59 pm)**
- More detailed instructions will be given on the course Canvas website, and discussed in class.

**Final letter grade assignments:**
Final letter grades will be determined from point percentages, using the standard ranges: A: 90 – 100; B: 80 - 89.9; C: 70 – 79.9; D: 60 – 69.9; R: < 60

**Late assignment policy:**
If you require extra time for an assignment, please get in contact with me as early as possible (email me with 85-418 or 85-814 in the subject line). Under normal circumstances, I will be able to grant a 48-hour extension up to twice during the semester.

**Academic integrity policy:**
Plagiarism and cheating are serious academic offenses with serious consequences. If you are discovered engaging in either behavior in this course, you will earn a failing grade on the assignment in question, and further disciplinary action may be taken.

For a clear description of what counts as plagiarism, cheating, and/or the use of unauthorized sources and tools, please see the University's Policy on Academic Integrity: https://www.cmu.edu/policies/student-and-student-life/academic-integrity.html

**Use of generative AI for assignments:**
Generative language models such as ChatGPT can rapidly generate text responses to prompts. In the course, we will talk in depth about language models and how they work, and we might even use tools like ChatGPT in the classroom. However, in this course it will not be acceptable to use these tools to generate your written assignments or presentations. It is permissible to use generative AI as you would a "search engine" to learn more about a topic, but your final work (sentences, paragraphs), must not be generated through these tools. If I suspect that your work has been written with the help of generative AI, I will ask you to meet with me to discuss it. This may lead to consequences similar to a plagiarism offense.

**Accommodations for students with disabilities:**
If you have a disability and require accommodations, please contact the Disability Resources Office (Catherine Getchell, Director of Disability Resources, 412-268-6121, getchell@cmu.edu). If you have an accommodations letter from the Disability Resources office, I encourage you to discuss your accommodations and needs with me as early in the semester as possible. I will work with you to ensure that accommodations are provided as appropriate.

**Inclusive excellence statement:**
In this course, we will have discussions on topics that span across academic disciplines, and we will each bring unique perspectives to these discussions. My goal this semester is that we will learn from each other in a variety of ways, and that we will foster an inclusive environment where we all feel empowered to share our perspective. If you experience any incident related to discrimination or bias, I encourage you to reach out to the Office for Institutional Equity and Title IX or make a confidential report.

**Statement on student wellness:**
It is common for us all to go through challenging periods, whether it be a problem with your mental health like anxiety or depression, substance use, relationship challenges, or stress from your life outside the classroom. These are very important issues and can impact your overall well-being as well as your performance in your courses. If you are struggling and need support, I am always here as a resource. There are also confidential mental health services available from Counseling and Psychological Services: https://www.cmu.edu/counseling/.

If mental health challenges are making it difficult for you to complete classwork, in this course or other courses, I also encourage you to reach out to the Office of Disability Resources to obtain the appropriate accommodations.

**Changes to the syllabus:**
After the start of the semester, I do not intend to make major changes to this syllabus, but it is possible that changes may be made. If there are changes to the schedule, assignments, or any other part of this syllabus, I will notify you either in class or via Canvas.